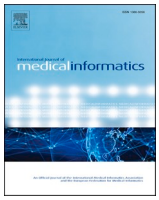
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Review article



Effects of digital parenting interventions on self-efficacy, social support, and depressive symptoms in the transition to parenthood: A systematic review and *meta*-analysis

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A R T I C L E I N F O

Keywords:
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A B S T R A C T

Background: Parenting self-efficacy is essential for the transition to parenthood. As digital parenting educational interventions are rapidly being developed, their effects have not been examined by pooling available randomized controlled trials (RCTs).

Objectives: To comprehensively investigate the effects of digital educational interventions on parents' self-efficacy, social support, and depressive symptoms in the first year after childbirth and identify the significant associated factors.

Methods: This study searched six electronic databases for relevant RCTs examining the efficacy of digital parenting interventions from inception to September 2022. The studied outcomes included changes in parent's self-efficacy, social support, and depressive symptoms observed after participating in a digital parenting program. The random-effects model was used to pool results. Subgroup and moderator analyses were performed.

Results: In total, seven RCTs enrolling 1342 participants were included. The parents who received digital parenting interventions had higher parenting self-efficacy (standardized mean difference [SMD]: 1.06, 95 % confidence interval [CI]: 0.40–1.71, p =.002) and social support (SMD: 2.72, 95 % CI: 0.38–5.07, p =.02) and decreased depressive symptoms at 3 months postpartum (SMD: −0.39, 95 % CI: −0.73 to − 0.04, p =.03). Providing the interventions for ≥ 6 weeks (SMD: 1.62, 95 % CI: 1.18–2.06, p <.001), providing in-person orientation (SMD: 1.88, 95 % CI: 1.32–2.44, p <.001), including a guided curriculum (SMD: 2.00, 95 % CI: 1.78–2.22, p <.001), and conducting interventions in Organisation for Economic Co-operation and Development countries (SMD: 1.98, 95 % CI: 1.78–2.19, p <.001) were identified as significant moderators.

Conclusions: Digital parenting interventions significantly increase parenting self-efficacy and social support as well as alleviate depressive symptoms for parents during their first year after childbirth. Such interventions can be beneficial for parents who prefer online education. Future studies investigating the long-term effects of these interventions are warranted.

Registration: The protocol for this systematic review and *meta*-analysis is registered in PROSPERO (registration number: CRD42021243641).

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1. Introduction

Enhancing self-efficacy in parenting considerably affects mothers' and fathers' mental well-being, children's development, and parent–child relationships, which are crucial for the smooth transition to parenthood [1,2]. Parents often experience the overwhelming process of providing infant care during their children's first year of life, with 25 % of mothers [3] and about 10 % of fathers [4] reporting low self-efficacy in caring for their newborns. Moreover, because nurturing a newborn is considerably stressful and challenging for both parents, they are prone to experiencing a lack of support or developing depressive symptoms [5,6]; an estimated 8.8 % of mothers and 2.6 % of fathers reported depression from prenatal to postpartum [7]. Research has shown that parents with high levels of parenting self-efficacy tend to experience less depression [8]. Consistent evidence highlights a positive association between parental confidence and social support levels [9]. Therefore, it is crucial to consider parental support and depression when investigating parenting self-efficacy. The recent trend towards shorter hospital stays after childbirth may overwhelm parents who need to acquire essential parenting skills in a limited time frame [10]. Consequently, ensuring the availability of accessible and continued care becomes vital in building parental confidence for new families experiencing early postnatal discharge [11]. This is particularly evident given the vast amount of information related to infant care and the rapid developmental changes that occur in infants [12]. The digital parenting approach has emerged as a potential solution amidst recent rapid advances in technology to support new parents' ongoing needs for learning parenting skills, alleviating emotional burdens, and facilitating role adaptation [13] during the child's first year of life.

A review reported that an educational intervention improved parents' confidence in their parenting ability; however, the effect of digital interventions remains unclear [14]. In the era of social media, parents can easily receive online peer support. Parents with online peer support have reported improved social support [15].The evidence of parenting interventions on depressive symptoms is mixed, with studies showing either a positive outcome [16] or no significant effect [17]. Given the recently published parenting studies that examine parental depression [18–21], a comprehensive synthesis of the current evidence on depression is needed. The inclusion of digital technologies in educational programs is recommended to improve parents' parenting knowledge and skills [22]. Health-care professionals can utilize new digital technology to address parental learning and emotional needs. However, evidence on the effectiveness of digital educational interventions is yet to be pooled. Whether the use of digital technologies in parent training programs can effectively support parents and reduce psychological distress remains unclear.

Digital educational interventions have gained popularity in recent years, especially among the younger generation [23], primarily because of their convenience, time flexibility, few geographical constraints, and high accessibility in providing parents with the required parenting competency [24]. In particular, digital interventions can offer innovative solutions by increasing parents' access to information, enhancing their adherence to program elements, and facilitating resources tailored to their specific context [25]. A well-designed digital educational curriculum would enable learners to achieve optimal learning outcomes, motivate interactions between learners and instructors, and build an online learning community [26]. With advances in educational technology, various online educational strategies, including audiovisual media, interactive media, and guided instruction, have become available and feasible for improving learning outcomes [27]. A well-structured digital program can help to tailor interventions, foster sufficient engagement in the dynamic parenting process, and achieve the expected impact [28]. However, the effects of these strategies used via digital parenting interventions on parenting outcomes are yet to be explored [29]. As technology resources vary greatly among countries, understanding the effects of digital educational interventions in different regions can provide educators with a crucial reference for developing culturally sensitive interventions. With the increasing use of digital educational interventions in health care currently, pooling the evidence on the features and effects of digital programs can help healthcare professionals meet the needs of parents during their transition to parenthood.

This systematic review and *meta*-analysis of randomized controlled trials (RCTs) investigated the effects of digital educational programs in enhancing parenting self-efficacy and social support, as well as alleviating depressive symptoms in the first year postpartum and identified the significant moderators of digital educational programs for parenting outcomes.

2. Material and methods

This systematic review and *meta*-analysis was conducted in accordance with Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [30], and was registered in the PROSPERO.

2.1. Search strategy

Six databases were searched for relevant studies: PubMed, Cochrane, CINAHL Plus, MEDLINE, EMBASE and Scopus. Two authors independently searched the databases from their inception to September 14, 2022.

The following Medical Subject Headings and text words were used to identify relevant studies: (Technology OR computer* OR web OR internet OR mobile* OR smartphone OR SMS OR video OR messag* OR application OR cellphone OR ios OR android OR Cell* OR telephone OR text OR online OR social media OR facebook OR youtube OR chat* OR Blogging) AND (pregnan* OR prenatal OR antenatal OR postnatal OR prepartum OR postpartum) AND (father* OR paternal OR mother* OR maternal OR parent*) AND (self-efficacy OR confidence OR competence OR esteem OR satisfaction OR self-regulation OR self-agency) AND (randomized controlled trial OR controlled clinical trial OR trial OR randomized OR randomly OR RCT OR random). In addition, the reference lists of included articles were manually searched for relevant studies.

2.2. Inclusion and exclusion criteria

Two authors (ML and CN) independently screened titles and abstracts to exclude irrelevant studies, and they then independently reviewed the full texts of potentially relevant articles. In this review, studies that met the following criteria were included: (1) those including pregnant or postnatal women and their partners, either primiparous or multiparous, who were parents (fathers or mothers) and had a healthy infant (aged between 0 and 12 months); (2) those examining the effects of digital interventions, such as internet-based, web-based, computer, and mobile applications. These types of digital interventions are widely used for health behavior [31], and have been recommended by the World Health Organization as scalable and accessible methods for promoting self-care [32]; (3) those including a comparison group who received active control or standardized usual care; (4) those examining the outcome measures of parenting self-efficacy by using validated tools, such as the Perceived Maternal Parental Self-efficacy (PMPSE) tool [33], Karitane Parenting Confidence Scale [34], and Maternal Confidence Questionnaire [35]; (5) the study design was a randomized controlled trial (RCT); and (6) those published in peer-reviewed journals. Randomized controlled trial (RCT) design was selected because it is considered as the gold standard for assessing the effectiveness of interventions [36]. No publication time or language restriction was applied. Studies were excluded if that included parents with mental health problems and mothers with high-risk pregnancy and were systematic reviews or *meta*-analyses of studies on parenting self-efficacy.

2.3. Data extraction and management

The two authors (ML and CN) used a standardized data extraction form to retrieve relevant data, including participant characteristics (e.g., country, sample size, and age), intervention characteristics (e.g., educational intervention timing, intervention provider, mode of delivery, duration, content, and theoretical framework), control group characteristics, and outcome measures (e.g., assessment tools and time points). Two reviewers verified the extracted data to ensure accuracy. Any discrepancies in the extracted data were resolved through discussion with a third author (SK).

2.4. Quality assessment

The two authors (ML and MH) independently reviewed each eligible study in accordance with the guidelines of *Cochrane Handbook for Systematic Reviews of Interventions* [37]. The Cochrane risk-of-bias tool for randomized trials was used to assess the risk of bias of the included studies, and the tool covers five domains of bias: (1) randomization process, (2) deviations from intended interventions, (3) missing outcome data, (4) measurement of the outcome, and (5) selection of the reported result. Each domain is rated as having"low risk," "some concerns," or "high risk" [37]. Any disagreements in the evaluation were resolved by seeking a third opinion if required (SK).

3. Methodological analysis

Meta-analyses were conducted to pool the data of similar outcomes by using Comprehensive Meta-Analysis Software, version 2 (Biostat, Englewood, NJ 2013). The heterogeneity of the included studies was examined using I^2; an I^2 value of > 50 % with significance ($p < 0.05$) indicated substantial heterogeneity [37].

3.1. Pooled effect estimates

The effect of the studied outcomes (parenting self-efficacy, social support, and depressive symptoms) was summarized using a random-effects model. The standardized mean difference (SMD) and 95 % confidence interval (CI) were calculated [37]. To examine the time effect of digital educational interventions on the studied outcomes, we pooled the data of the included studies and reported the results at 1, 3, and 6 months postpartum.

3.2. Subgroup and meta-regression analyses

Subgroup analyses were conducted by dividing studies into groups based on intervention characteristics and methodological features. The types of participants, intervention timing, duration, intervention modes, key strategies, countries, and risk of bias were included in this analysis to identify the potential factors associated with the outcomes [37]. Meta-regression analyses were performed to examine the moderator effects of continuous variables, such as age.

3.3. Publication bias

The current study assessed publication bias by using Begg and Mazumdar's rank correlation test, with a p-value of < 0.05 indicating the presence of publication bias [38]. Duval and Tweedie's trim and fill method was used to adjust for publication bias by comparing pooled estimates after considering the number of unpublished studies in the analysis [39].

4. Results

4.1. Selection of studies

In total, 6063 studies were identified from the six electronic databases. Among them, 3686 duplicate articles were excluded. The titles and abstracts of the remaining 2377 articles were screened, and 2246 studies were excluded because they did not meet the inclusion criteria. After the evaluation of the full text of the articles (n = 131), 124 articles were excluded as they did not include parenting self-efficacy data, included parents with mental health problems, did not examine digital educational interventions, or were not designed as an RCT. Finally we included seven studies in this *meta*-analysis, with seven articles examining the parenting self-efficacy outcomes [18–21,40–42], five articles assessing the social support outcome [18–21,42], and four articles reporting the depressive symptoms of parents [18–21] (Fig. 1).

4.2. Characteristics of included studies

Table 1 lists the characteristics of the included studies. Each RCT included parents who received digital educational interventions (the intervention group) or usual care (the control group). These studies were published between June 2009 and April 2021. Most programs were offered in countries that were not part of the Organisation for Economic Co-operation and Development (OECD; n = 5), namely Singapore, China, and Taiwan. In total, 1342 parents were included in this systematic review and *meta*-analysis. The mean age of the parents ranged from 27.3 to 32.7 years. Most studies included only mothers (n = 5), whereas two studies included both mothers and fathers. The providers of all interventions were either nurses or midwives, and most of the interventions were delivered during the postnatal period (n = 4).

Digital educational interventions were offered through a website (n = 5) or mobile application (n = 2). The parents received the following content of the intervention: infant growth and development, daily infant care, infant feeding, common illnesses in infants, postnatal care for women, social support, parent–infant relationship, father's role, and transition into parenthood. The control group received routine care including routine health care and education at the clinic, during hospitalization, during home visits, or at hospital/clinic follow-up. The intervention duration ranged from 1 to 6 months; the intervention duration of four studies was $\geq$ 6 weeks.

Most studies (n = 6) used Bandura's self-efficacy theory as the theoretical framework [43]. Three studies included interventions based on a combination of social exchange theory [44], Pender's Health Promotion Model [45], or Bowlby's theory of attachment [46]. All the included studies used validated self-reported tools to examine parenting self-efficacy (e.g., PMPSE scale) [33], social support (e.g., Postnatal Social Support Scale) [47], and depressive symptoms (e.g., Edinburgh Postnatal Depression Scale) [48]. The follow-up period of parenting self-efficacy ranged from $\leq$ 1 month (n = 5) or 6 weeks to 3 months postpartum (n = 5) and 6–9 months postpartum (n = 3; Table 1).

4.3. Key strategies used in digital interventions

All digital interventions used a web design for encouraging interactions between learners and instructors as well as among learners (e.g., group chats, group forums, and questions asked with responses received from experts). Most programs used various measures to improve the participants' adherence, including maintaining contact (through email, message, and phone calls) and monitoring the frequency at which the participants logged into the website or mobile application (Table 2). All digital interventions used a common digital delivery method that combined video and text, and some also included audio materials (n = 3). Each method, either in-person orientation or guided curriculum, was used by only one study respectively (n = 1).

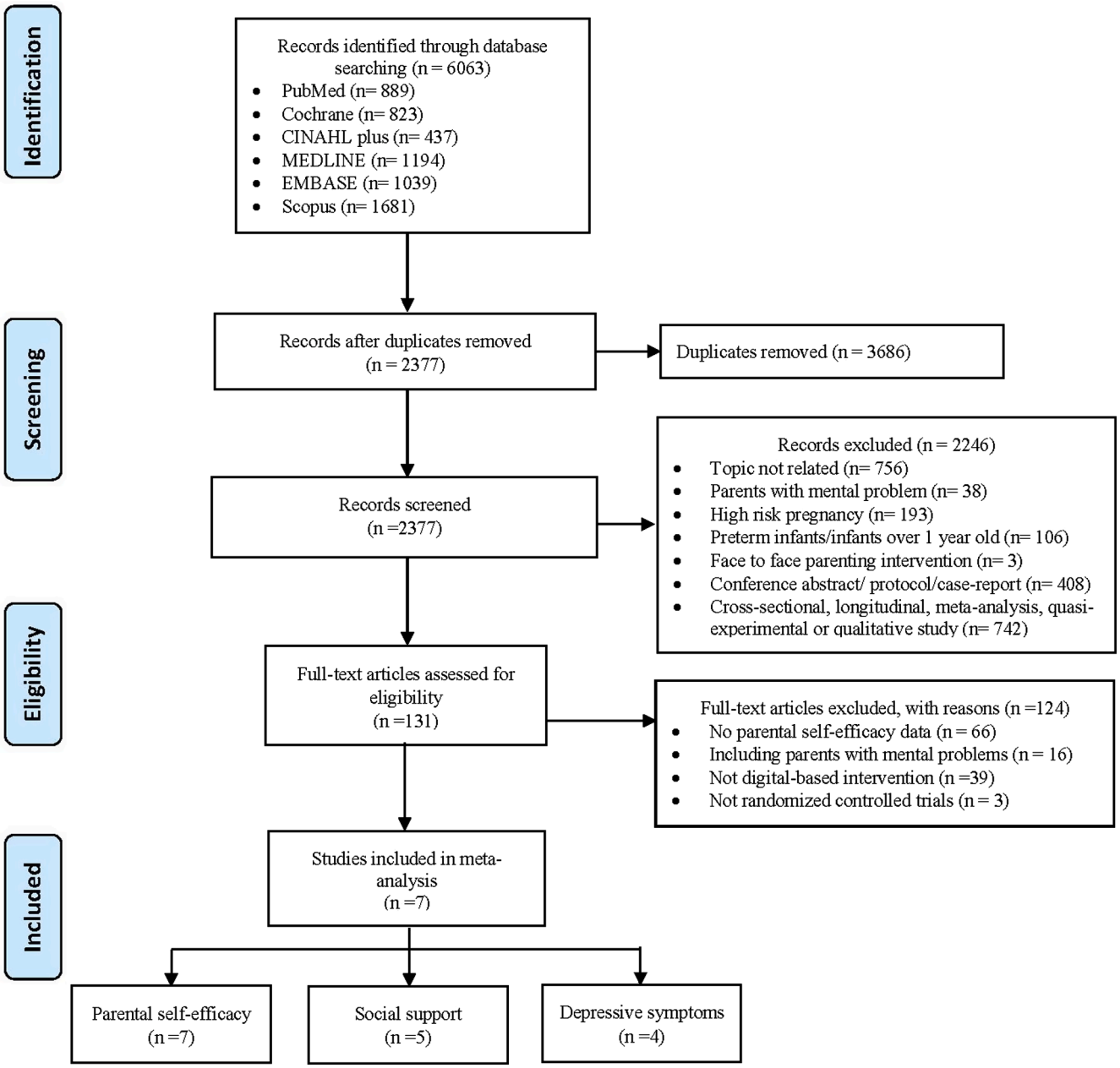


Fig. 1. PRISMA flow diagram.

4.4. Quality assessment of included studies

The results of the overall risk of bias indicated that most studies had low to high risk of bias. Three studies were classified as low risk, two were classified as some concern, and two were classified as high risk. The studies with a high risk of bias were due to they have high rates of missing data (Supplementary Table 1).

4.5. Effects of digital educational interventions on parenting self-efficacy, social support, and depressive symptoms

The results of Cochran's Q test revealed high heterogeneity (Cochran's Q test, p <.001; I^2 = 96 %), indicating a large degree of differences in the effect sizes of the included studies. A random-effects model was adopted for pooling the SMD of parenting self-efficacy for digital educational interventions (SMD: 1.06, 95 % CI: 0.40–1.71; p =.002), and the results showed that digital educational interventions improved parenting self-efficacy. Compared with the control group, the intervention group had higher parenting self-efficacy at postpartum ≤ 1 month (SMD: 0.65, 95 % CI: 0.13–1.17) and during postpartum 6 weeks to 3 months (SMD: 0.99, 95 % CI: 0.18–1.79), but not at 6–9 months postpartum (SMD: 1.01, 95 % CI: −0.33–2.34; Table 3).

Regarding social support, the intervention group exhibited a significantly higher level of social support (SMD: 2.72, 95 % CI: 0.38–5.07; p =.02) than did the control group. In particular, the effect of the higher level of social support was maintained at 6 weeks to 3 months postpartum (SMD: 0.53, 95 % CI: 0.15–0.91). No significant difference in depressive symptoms was observed in the pooling effect between the intervention and control groups (SMD: −0.22, 95 % CI: −0.47–0.04). However, a significant difference in depressive symptoms was observed at 6 weeks to 3 months postpartum between the groups (SMD: −0.39, 95 % CI: −0.73 to − 0.04; Table 3).

Table 1
Characteristics of study participants and interventions.

Author (year)/ Country	Participants	Intervention group Intervention timing/ Provider	Intervention Modes/ Duration	Contents	Control group	Theoretical framework	Outcome measures: tools/Time points
Huang et al. (2021), China [18]	- N = 40 mothers - Mean age, 27.3 years	Postnatal/ Nurse	Web-based/> 3 months	- Growth and development - Daily infant care - Breastfeeding and bottle feeding - Common illness for infants - Postnatal care for women	Routine care: routine education at hospitalHome visits	Bandura's self-efficacy theory and social exchange theory	Outcomes: - Parenting self-efficacy: SICS - Postnatal depression: EPDS - Social support: PSSS Time points: Baseline and 3 and 6 months postpartum
Jiao et al. (2019), Singapore [19]	- N = 136 mothers - Mean age. 30.7 years	Postnatal/ Nurse	Web-based/1 month	- Daily infant care - Postnatal care for women - Social support	Routine care: routine education at hospitalHospital follow-up	Bandura's self-efficacy theory	Outcomes: - Parenting self-efficacy: PMPSE - Postnatal depression: EPDS - Social Support: PICSS-modified Time points: Baseline and 1, 3, and 6 months postpartum
Kuo et al. (2009), Taiwan [40]	- N = 118 mothers - Mean age, 30.1 years	Prenatal, Postnatal/ Nurse	Web-based/3 months	- Daily infant care - Breastfeeding - Parent–infant relationship - Father's role	Routine care: routine education at hospital	Bandura's self-efficacy theory	Outcomes: - Parenting self-efficacy: MCQ Time points: 2 and 6 weeks postpartum
Sari and Altay (2020), Turkey [41]	- N = 71 mothers - Mean age. 27.5 years	Prenatal/Nurse	Web-based/1.5 months	- Daily infant care - Breastfeeding - Parent–infant relationship	Routine care: routine education at clinicClinic follow-up	Pender's Health Promotion Model + self-efficacy	Outcomes: - Parenting self-efficacy: PSS Time points: 1 weeks and 3 months postpartum
Sawyer et al. (2017), Australia [42]	- N = 491 mothers - Mean age. 32.5 years	Postnatal/ Nurse	Web-based/> 6 months	- Growth and development - Daily infant care - Parent–infant relationship - Postnatal care for women	Home visits	NR	Outcomes: - Parenting self-efficacy: KPCS Social support: MSS Timepoints: Baseline and 9, 15, and 21 months postpartum
Shorey et al. (2017), Singapore [20]	- N = 250 mothers and fathers - Mean age, 32.7 years	Postnatal/ Midwife	Mobile application /1 month	- Daily infant care - Breastfeeding - Postnatal care for women - Father's role	Routine care: routine education at hospitalHospital follow-up	Bandura's self-efficacy theory	Outcomes: - Parenting self-efficacy: PES - Postnatal depression: EPDS Social support: PSSP Time points: Baseline and 1 month postpartum
Shorey et al. (2019), Singapore [21]	- N = 236 mothers and fathers - Mean age, 32.0 years	Prenatal, Postnatal/ Midwife	Mobile application /1 month	- Daily infant care - Breastfeeding - Postnatal care for women - Parent-infant relationship - Transition into parenthood	Routine care: routine education at hospitalHospital follow-up	Bandura's self-efficacy theory and Bowlby's theory ofattachment	Outcomes: - Parenting self-efficacy: PES - Postnatal depression: EPDS - Social support: PSSP Time points: Baseline and 2 days, 1 month, and 3 months postpartum

Note: RCT, randomized controlled trial; SICS, Self-efficacy in Infant Care Scale; PMPSE, Perceived Maternal Parental Self-efficacy Scale; MCQ, Maternal Confidence Questionnaire; PSS, Parental Self-efficacy Scale; KPCS, Karitane Parenting Confidence Scale; PSE, Parenting Self-efficacy; EPDS, Edinburgh Postnatal Depression Scale; PSSS, Postnatal Social Support Scale; PICSS-modified, modified Perinatal Infant Care Social Support tool; MSS, Maternal Support Scale; PSSP, Perceived Social Support for Parenting; NR: not reported.

4.6. Moderator analysis

To determine the factors affecting parenting self-efficacy and social support, a moderator analysis was conducted. The results of the subgroup analysis indicated that the parents who received digital educational interventions for $\geq$ 6 weeks had significantly higher parenting self-efficacy scores than did those who received digital educational interventions for < 6 weeks (SMD: 1.62, 95 % CI: 1.18–2.06,

Table 2
Classification of the key strategy used in digital educational interventions.

Study	Digital delivery method audio	video	text	In-person orientation	Guided curriculum	Interactive activity	Adherence measure
Huang et al. (2021) [18]	–	+	+	–	–	+	+
Jiao et al. (2019) [19]	+	+	+	–	–	+	+
Kuo et al. (2009) [40]	–	+	+	–	–	+	+
Sari and Altay (2020) [41]	–	+	+	+	–	+	+
Sawyer et al. (2017) [42]	–	+	+	–	+	+	+
Shorey et al. (2017) [20]	+	+	+	–	–	+	+
Shorey et al. (2019) [21]	+	+	+	–	–	+	+

Note: +, present; -, absent.

Table 3
Overall effect of digital educational interventions on parenting self-efficacy, social support, and depressive symptoms.

Outcome	*N*	Overall effect SMD [95 % CI]	p value	Heterogeneity I^2	p
Parental self-efficacy	7	1.06 [0.40–1.71]	0.002	96 %	< 0.001
≤ 1 month postpartum	5	0.65 [0.13–1.17]	0.014		
6 weeks- 3 months postpartum	5	0.99 [0.18–1.79]	0.02		
6–9 months postpartum	3	1.01 [-0.33–2.34]	0.14		
Social support	5	2.72 [0.38–5.07]	0.02	99 %	< 0.001
≤ 1 month postpartum	3	0.52 [-0.07–1.11]	0.09		
6 weeks-3 months postpartum	3	0.53 [0.15–0.91]	0.006		
6–9 months postpartum	3	3.75 [-2.43–9.93]	0.24		
Depressive symptoms	4	−0.22 [-0.47–0.04]	0.10	58 %	0.07
≤ 1 month postpartum	3	−0.12 [-0.28–0.03]	0.13		
6 weeks-3 months postpartum	3	−0.39 [-0.73 to −0.04]	0.03		
6–9 months postpartum	2	−0.49 [-1.42–0.44]	0.30		

Note: *N*, number of studies; CI, confidence interval; SMD, standardized mean difference. I^2, variation across studies due to heterogeneity.

p <.001). In addition, intervention program that included in-person orientation resulted in significantly higher parenting self-efficacy scores than did those that did not offer in-person orientation (SMD: 1.88, 95 % CI: 1.32–2.44, p <.001; Table 4). Moreover, the parents who received a guided curriculum exhibited higher parenting self-efficacy (SMD: 2.00, 95 % CI: 1.78–2.22, p <.001) and social support (SMD: 10.9, 95 % CI: 10.20–11.61, p <.001) than did those who receive the unguided curriculum.

The parents from OECD countries were more likely to have a higher parenting self-efficacy score (SMD: 1.98, 95 % CI: 1.78–2.19, p <.001) and higher social support (SMD: 10.9, 95 % CI: 10.20–11.61, p <.001) than did the parents from non-OECD countries (Table 4). The results did not detect significant differences in the following factors: the type of participants (i.e., both mothers and fathers or mothers only), intervention timing, mode of intervention, risk of bias, or age of participants.

4.7. Publication bias

The results of Begg and Mazumdar's rank correlation test (Tau = 0.14, p =.65) indicated no significant publication bias. The overall effect remained the same after using Duval and Tweedie's trim and fill method.

5. Discussion

This systematic review and *meta*-analysis demonstrated the positive effects of digital educational interventions on parenting outcomes and the associated factors. To the best of our knowledge, this is the first systematic review and *meta*-analysis of RCTs examining the effects of digital educational interventions. The results revealed that digital educational interventions effectively increased parents' self-efficacy and social support and reduced their depressive symptoms from 6 weeks to 3 months postpartum. Interventions that were provided for ≥ 6 weeks, included in-person orientation, followed a guided curriculum, and were conducted in OECD countries, significantly influenced the effects of the intervention. However, no significant differences were found in the study participants (whether including both mothers and fathers or only mothers), the age of the participants, the timing of the intervention, the mode of intervention, or the levels of risk of bias. The results of the *meta*-analysis indicated that digital parenting programs can enhance parents' confidence and social support while also potentially lessening their depressive symptoms. Thus, digital parenting programs can be effective alternatives to face-to-face parenting programs in this digital era.

This study demonstrated that digital educational interventions effectively increased parenting self-efficacy. The significant effect was evident following childbirth through 3 months postpartum, but it was not observed at 6–9 months postpartum. A previous *meta*-analysis reported that face-to-face parenting programs increased parenting self-efficacy in first-time parents (SMD: 0.57, 95 % CI: 0.44–0.70) [14], with their effectiveness lower than that of the digital parenting interventions included in the current review (SMD: 1.06, 95 % CI: 0.40–1.71). However, the current review included only a small number of studies with limited follow-up time; thus, the long-term effect of digital interventions could not be examined. Additional studies, with a follow-up period of more than 9 months, are needed to investigate the sustained effect of digital parenting programs.

Most studies included in this review employed theory-based programs that adopted self-efficacy theory [43,49] to improve parenting self-efficacy, indicating the beneficial effect of theory-based programs on parenting outcomes. In line with previous *meta*-analysis of face-to-face parenting programs, intervention that incorporate the four sources of parenting self-efficacy from Bandura's theory– namely, performance achievement, vicarious experiences, verbal persuasion, and physiological states were effective in boosting parental confidence [14]. Our study results indicated that combining digital technology with theory-based programs enabled parents to gain parenting knowledge and skills and receive digital feedback from health-care providers [50]. In parenting research, theoretical frameworks play an important role in guiding the development and implementation of effective educational programs with favorable outcomes [51]. Adopting such theoretical frameworks is advantageous for health care providers to incorporate digital elements into intervention programs in order to achieve the expected outcomes. For example, providing online educational resources supports the performance achievement component, facilitating online interactions enhances vicarious experiences, and tracking engagement levels can improve adherence [13]. Theory-based approaches to digital parenting interventions provide an evidence-based foundation and effective strategies that can better ensure desired learning outcomes [52]. The development of future digital parenting programs should translate theory into practice that will meet parents' needs, engage their

Table 4
Moderator analysis of digital interventions on parenting self-efficacy and social support.

Variables	Parenting self-efficacy			Social support		
	N	SMD [95 % CI]	p	*N*	SMD [95 % CI]	p
Participants			0.19			0.28
Mothers and fathers	2	0.50 [−0.36–1.37]		2	0.65 [−0.23–1.52]	
Mothers only	5	1.29 [0.50–2.08]		3	4.14 [−2.15–10.42]	
Intervention timing			0.12			0.07
Prenatal period only	1	1.88 [1.32–2.44]		0		
Postnatal period only	4	1.03 [0.16–1.89]		4	3.37 [−0.02–6.76]	
Prenatal and postnatal period	2	0.73 [-0.60–2.06]		1	0.20 [−0.05–0.46]	
Duration			0.001			0.25
Less than 6 weeks	3	0.40 [−0.17–0.97]		3	0.52 [−0.07–1.11]	
6 weeks and more	4	1.62 [1.18–2.06]		2	6.08 [−3.37–15.54]	
Intervention modes			0.19			0.28
Mobile application	2	0.50 [−0.36–1.37]		2	0.64 [−0.23–1.52]	
Web-based	5	1.29 [0.50–2.08]		3	4.14 [−2.15–10.42]	
In-person orientation			0.04			–
No	6	0.93 [0.21–1.64]		5	–	
Yes	1	1.88 [1.32–2.44]		0	–	
Guided curriculum			0.001			0.001
No	6	0.88 [0.34–1.42]		4	0.66 [0.13–1.20]	
Yes	1	2.00 [1.78–2.22]		1	10.9 [10.20–11.61]	
Countries			0.001			0.001
OECD	2	1.98 [1.78–2.19]		1	10.9 [10.20–11.61]	
non-OECD	5	0.69 [0.17–1.22]		4	0.66 [0.13–1.20]	
Risk of bias			0.054			0.13
LR	3	1.05 [0.02–2.08]		3	4.07 [−0.20–8.34]	
SC	2	0.45 [−0.41–1.31]		2	0.68 [−0.36–1.71]	
HR	2	1.61 [1.17–2.05]		0		
Age (year)	7	$\beta = -0.073$	0.67	5	$\beta = -0.665$	0.31

Note: *N*, number of studies; CI, confidence interval; SMD, standardized mean difference; OECD, Organisation for Economic Co-operation and Development; LR, Low risk; SC, Some concern; HR, High risk.

participation, and increase their confidence in parenting [53].

Digital parenting interventions can provide parents with professional support by offering scientific knowledge from video–audio media or text-based materials and by enabling the parents to interact with other parents through group chats or discussion forums. In this *meta*-analysis, the intervention group reported a higher level of social support than did the control group, with a significant effect observed at 3 months postpartum. A previous study reported the usefulness of online parental interventions or social media in providing professional support and peer support to parents [54]. Digital interventions provide flexibility by enabling parents to raise their concerns and receive timely professional feedback, thus supporting and addressing parents' needs efficiently [23]. Digital parental interventions can enhance parenting confidence by offering information and peer support, thus improving their overall social support.

This *meta*-analysis revealed a significant decrease in depressive symptoms at 6 weeks to 3 months postpartum; however, this decrease was not observed at later time points after digital parenting interventions. This finding indicated that digital parenting interventions aimed at improving parenting skills have less coverage for the emotional aspect. A previous study reported that digital mental health interventions effectively reduced depressive symptoms in postpartum women (SMD: −0.16, 95 % CI: −0.26 to −0.05) [55] and improved depressed mood in parents of young children (SMD: 0.38, 95 % CI: 0.17–0.59) [56]. Although the course content differs between parenting and mental health programs, the level of improvement in depressive symptoms following both programs was similar. The association between parenting self-efficacy and depressive symptoms might be attributable to the effect of a digital parenting program on improving depressive symptoms [57]. Future self-efficacy parenting programs should consider integrating emotional coping training to help parents with depressive symptoms.

The results revealed that the parents who received digital educational interventions for ≥ 6 weeks experienced an improvement in their parenting self-efficacy. A previous study on face-to-face parenting programs reported that participants' self-efficacy was significantly increased for programs with a duration of ≥ 10 weeks [14]. Although the optimal duration varies, both digital and face-to-face parenting approaches suggest that longer programs are beneficial for new parents. This finding reflects the fact that sufficient time and resources are helpful for parents to transition into their new roles and become confident in infant care practices [58]. Several advantages of digital educational interventions, including easy access to learning, the ability to repeatedly view learning materials, or provision of feedback by instructors, might improve the flexibility and effectiveness of digital learning in a reasonable duration [59]. Notably, significant intervention effects were found for all outcomes between 6 weeks and 3 months postpartum, but not before 4 weeks for social support and depression. Social learning theory suggests that both formal and informal support play a crucial role in new parents' adjustment to parenthood [43]. In cultures with a postpartum confinement tradition [60] or countries with supportive family leave policies [61], new parents may receive enhanced comprehensive support following childbirth, which may attenuate the early effects of parental intervention programs. Due to the limited studies on long-term effects, the non-significant findings at 6–9 months postpartum were less conclusive. Future research should investigate the persistence of program effects.

The subgroup analysis indicated that programs incorporating coaching, including in-person orientation or a guided curriculum, effectively increased parenting self-efficacy; this finding is consistent with those of previous studies combining face-to-face sessions with online sessions [62]. During in-person orientation, participants have the opportunity to overview the entire program structure and usage and receive instant feedback if required. These features are crucial for individuals unfamiliar with the user interface of e-learning platforms [63]. The current study demonstrated the positive effect of using a guided program, corroborating previous findings that individuals participating in a guided program coped with stress more effectively (Cohen's d: 0.64, 95 % CI: 0.50–0.79) [64]. Together, these results indicate that a hybrid model integrating in-person components (i.e., in-person orientation or a guided curriculum) into a digital mode can effectively increase parents' confidence and social support.

This study demonstrated that educational programs provided in OECD countries improved parenting self-efficacy and social support. Digital health resources are becoming valuable for health promotion in OECD countries [65]. Digital users in OECD countries exhibited a higher level of satisfaction with and acceptance of digital health resources [66]. Parents in OECD countries might be more receptive to digital educational programs. The knowledge, skills, and assistance provided by healthcare professionals can motivate parents to improve their self-efficacy and social support.

Most studies included in this review examined only mothers' self-efficacy and did not focus on fathers in the parenting program. In recent years, cumulative evidence has emphasized the crucial role of the father in parental support and the beneficial effects of fathers' involvement on children's development [67]. Future parenting programs focusing on both mothers and fathers are warranted to investigate the effect of father engagement on parent self-efficacy and support.

This study has several limitations that should be considered when interpreting the results. First, the included studies examined parenting self-efficacy by using a commonly used parent-reported questionnaire; thus, the presence of self-reporting bias cannot be ruled out. Second, high heterogeneity was observed in the included studies. Subgroup and *meta*-regression analyses were conducted to determine the crucial factors underlying heterogeneity. Finally, the included studies were from China, Singapore, Taiwan, Turkey, and Australia, there may be limited discussion on differences in geographical region. However, there were no language restrictions, or publication bias detected in this review.

6. Conclusions

Digital educational programs can significantly increase parenting self-efficacy and social support and reduce depressive symptoms from 6 weeks to 3 months postpartum. In the future, health-care professionals should consider employing digital programs to provide parents easy access to health promotion and educational interventions at a reasonable cost for improving family health. Additional studies should investigate whether digital educational programs can enhance parenting self-efficacy and social support and alleviate depressive symptoms over the long term, that is, after 1 year postpartum.

7. Summary points

What is already known about the topic?

- During the first year after a child's birth, parents commonly encounter stressful and challenging tasks in parenting, which often lead to increased depressive symptoms and a lack of social support.
- Educational interventions during the perinatal period can improve parenting self-efficacy, which is closely linked to positive parenting and child development.
- Digital parenting educational interventions are being rapidly developed; however, their effects and associated factors have not been confirmed by pooling available randomized controlled trials.

What this study added to our knowledge?

- Digital parenting educational interventions effectively increased parenting self-efficacy scores and social support and alleviated depressive symptoms at 3 months postpartum.
- Providing interventions for $\geq$ 6 weeks, providing in-person orientation, including a guided curriculum, and conducting interventions in Organisation for Economic Co-operation and Development countries were significant factors associated with increased levels of parenting self-efficacy or social support.
- Healthcare professionals should consider digital programs as an effective alternative channel to deliver educational interventions for improving parental and family health.

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CRediT authorship contribution statement

Marianne Lin-Lewry: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft. **Cai Thi Thuy Nguyen:** Software, Visualization, Writing – review & editing. **Mega Hasanul Huda:** Software, Visualization, Writing – review & editing. **Shao-Yu Tsai:** Investigation, Visualization, Writing – review & editing. **Roselyn Chipojola:** Visualization, Writing – review & editing. **Shu-Yu Kuo:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Resources, Software, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijmedinf.2024.105405.

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